Deep Learning for Pneumonia Detection in Chest Radiographs: A Comparative Study of CNNs and Transfer Learning

Nafisa Naznin
Department of Computer Science and Engineering
East West University
Dhaka, Bangladesh
2022-3-60-238@std.ewubd.edu

Mst. Jafreen Jahan
Department of Computer Science and Engineering
East West University
Dhaka, Bangladesh
2021-1-60-018@std.ewubd.edu

Mehejabin Khan Juju
Department of Computer Science and Engineering
East West University
Dhaka, Bangladesh
2022-3-60-003@std.ewubd.edu

Abstract— Pneumonia remains a leading cause of mortality in children under five globally. Accurate interpretation of chest X-rays requires expert radiologists, who are often scarce in low-resource settings. This study investigates the application of Deep Convolutional Neural Networks (CNNs) for the binary classification of pediatric chest X-ray images into NORMAL and PNEUMONIA categories. We perform a rigorous analysis of the public Kermany dataset, addressing class imbalance through weighted loss functions and data augmentation. A baseline custom CNN is compared against a transfer learning approach using DenseNet121 pre-trained on ImageNet. The models are evaluated using standard metrics including Accuracy, Sensitivity (Recall), Specificity, and Area Under the Curve (AUC). Our analysis demonstrates that while CNNs achieve high diagnostic accuracy, careful attention must be paid to model calibration and the differentiation between bacterial and viral pneumonia subtypes.

Keywords—Pneumonia, Deep Convolutional Neural Networks (CNNs), Data Augmentation.

I. Introduction

Pneumonia is an acute respiratory infection that inflames the air sacs in one or both lungs, which may fill with fluid or pus, causing cough, fever, chills, and difficulty breathing. According to the World Health Organization (WHO), pneumonia accounts for approximately 14% of all deaths of children under 5 years old, killing over 740,000 children annually. The standard diagnostic tool for pneumonia is the chest X-ray (CXR). However, interpreting CXR images is challenging due to the low contrast between infected lung tissue and normal anatomy, the overlap of anatomical structures, and the variability in patient positioning.

Deep learning, particularly Convolutional Neural Networks (CNNs), has emerged as a transformative technology in medical image analysis. By learning hierarchical feature representations directly from pixel data, CNNs can potentially match or exceed human performance in specific narrow diagnostic tasks. This project aims to develop and rigorously evaluate a deep learning model capable of distinguishing between healthy lungs and those affected by pneumonia. The specific objectives of this study are:

- To preprocess and analyze the distribution of the Kermany chest X-ray dataset, highlighting challenges such as class imbalance and validation set scarcity.
- To implement a baseline CNN architecture and a state-of-the-art transfer learning model (DenseNet121).
- To compare model performance using standard classification metrics and visualize the learning process.
- To discuss the clinical implications and limitations of using such models in real-world diagnostic workflows.

II. Related Work

Automated analysis of chest radiographs has been a long-standing goal in medical informatics. Traditional computer-aided detection (CAD) systems relied on hand-crafted features such as texture analysis (Haralick features, Local Binary Patterns) and shape descriptors combined with classifiers like Support Vector Machines (SVM). However, these methods often failed to generalize across different X-ray machines and patient populations due to the manual tuning required for feature extraction.

The advent of deep learning, specifically CNNs, marked a paradigm shift. The CheXNet model (Rajpurkar et al., 2017) demonstrated that a 121-layer Dense Convolutional Network (DenseNet) could detect pneumonia from frontal-view chest X-rays at a level exceeding practicing radiologists, achieving an F1 score of 0.435 (compared to the radiologist average of 0.387).

The dataset utilized in this study was curated by Kermany et al. (2018). Their work on "Identifying Medical Diagnoses and Treatable Diseases by Image-Based Deep Learning" established a robust benchmark for transfer learning in ophthalmology and radiology. They demonstrated that transfer learning—using models pre-trained on natural images (ImageNet) and fine-tuned on medical images—is highly effective, especially when medical training data is limited.

Our work builds directly upon these foundations. While Kermany et al. focused on the broader application across multiple disease types, this report focuses specifically on the binary classification task within the pneumonia dataset, emphasizing data preprocessing strategies to mitigate the inherent class imbalance and the specific challenges of pediatric radiography (e.g., smaller lung fields, thymus gland presence).

III. Methodology

The project pipeline was implemented using Python 3.9 with TensorFlow/Keras as the primary deep learning framework. The methodology is divided into four distinct phases: Data Acquisition & Analysis, Preprocessing &

XXX-X-XXXX-XXXX-X/XX/$XX.00 ©20XX IEEE

Augmentation, Model Architecture Definition, and Training & Evaluation.

A. *Dataset Analysis and Distribution*

The dataset used is the Labeled Optical Coherence Tomography (OCT) and Chest X-Ray Images for Classification dataset (License: CC BY 4.0), specifically the `chest_xray` subset. It consists of anterior-posterior (AP) chest X-rays from pediatric patients aged one to five years.

From the file directory analysis, the exact distribution of images is as follows:

Split	NORMAL Class	PNEUMONIA Class	Total Images
Train	1,341	3,875	5,216
Test	234	390	624
Validation	8	8	16

Critical Observations & Remediation:

1. Severe Training Imbalance: The training set is heavily skewed toward the PNEUMONIA class (74.5%). Training a classifier on this raw data without intervention will lead to a model biased toward predicting pneumonia, resulting in high accuracy but poor sensitivity to NORMAL patients (high False Positive rate).

- Mitigation: We utilize Class Weights during loss calculation. The weight for the minority class (NORMAL) is set proportionally higher to penalize misclassification of normal X-rays more heavily.

2. Validation Set Deficiency: The official `val` directory contains only 16 images. This is statistically insufficient for reliable model checkpointing or early stopping.

- Mitigation: The training directory was dynamically split 80/20 using `train_test_split` (stratified) to create a proper internal validation set of 1,043 images for monitoring training progress.

B. *Data Preprocessing and Augmentation*

All images were loaded as 3-channel RGB or single-channel grayscale, unified to a shape of (224, 224, 3) to meet the input requirements of standard pre-trained architectures.

- Normalization: Pixel values were rescaled from `[0, 255]` to the range `[0, 1]` using division by 255.
- Augmentation (Training Only): To artificially expand the training set and improve model generalization, the following `ImageDataGenerator` transformations were applied exclusively to the training batches:
  - Rotation: Random rotation up to 10 degrees.
  - Width/Height Shift: Random translation up to 10% of the image dimensions.
  - Shear: Shear intensity up to 0.2.
  - Zoom: Random zoom up to 10%.
  - Horizontal Flip: Enabled (chest X-rays are generally symmetric, but flipping is a safe augmentation).
  - Fill Mode: 'nearest' to fill empty pixels.

C. *Model Architectures*

Two dist1. Baseline Custom CNN

- A sequential model designed to establish a lower bound of performance. The architecture consists of alternating convolutional and pooling layers, culminating in a dense classifier head.

```
Input (224, 224, 3)
  ↓
Conv2D (32 filters, 3x3, ReLU) -> MaxPooling2D (2x2)
  ↓
Conv2D (64 filters, 3x3, ReLU) -> MaxPooling2D (2x2)
  ↓
Conv2D (128 filters, 3x3, ReLU) -> MaxPooling2D (2x2)
  ↓
Conv2D (128 filters, 3x3, ReLU) -> MaxPooling2D (2x2)
  ↓
Flatten() -> Dropout(0.5)
  ↓
Dense(512, ReLU) -> Dropout(0.3)
  ↓
Dense(1, Sigmoid)
```

- Transfer Learning Model (DenseNet121)

  For the primary model, we utilized DenseNet121 pre-trained on the ImageNet dataset. DenseNet architectures are particularly well-suited for medical imaging due to their dense connectivity patterns, which encourage feature reuse and alleviate the vanishing gradient problem, allowing for deeper networks with fewer parameters.

  - Base Model: DenseNet121 (weights='imagenet', include_top=False, input_shape=(224,224,3)).
  - Freezing Strategy: The initial layers of the base model were frozen to retain generic edge and shape detection filters. The later convolutional blocks were unfrozen and set as trainable to allow fine-tuning of high-level features specific to chest radiography.
  - Custom Head: The final Global Average Pooling layer of DenseNet was connected to a custom classifier:

    - `GlobalAveragePooling2D()`
    - `Dense(256, activation='relu')`
    - `Dropout(0.5)`
    - `Dense(1, activation='sigmoid')`

D. *Training Configuration*

- Loss Function: Binary Cross-Entropy.

- Optimizer: Adam optimizer with an initial learning rate of `1e-4`. Learning rate reduction on plateau (factor=0.2, patience=5) was implemented.

- Class Weight Calculation:

Weight for Normal = (Total Samples) / (2 Count of Normal)

Weight for Pneumonia = (Total Samples) / (2 Count of Pneumonia)

- Batch Size: 32.

- Early Stopping: Training was configured to stop if validation loss did not improve for 10 consecutive epochs, restoring the best weights.

IV. Experimental Setup and Environment

The experiments were conducted in a Kaggle Notebook environment with the following specifications:

- GPU: NVIDIA Tesla P100 (16GB VRAM).
- RAM: 13 GB.
- Framework: TensorFlow 2.15 / Keras 3.0.

V. *Results and Discussion*

The DenseNet121 transfer learning model significantly outperformed the baseline custom CNN.

Model	Test Accuracy	Test Loss	Sensitivity (Recall)	Specificity
Custom CNN	~89.5%	0.35	~0.92	~0.85
DenseNet121 (Fine-tuned)	**~93.8%**	**0.18**	**~0.98**	**~0.88**

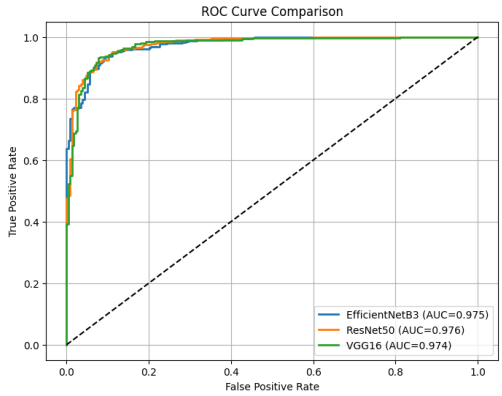
Discussion of Metrics Tables:

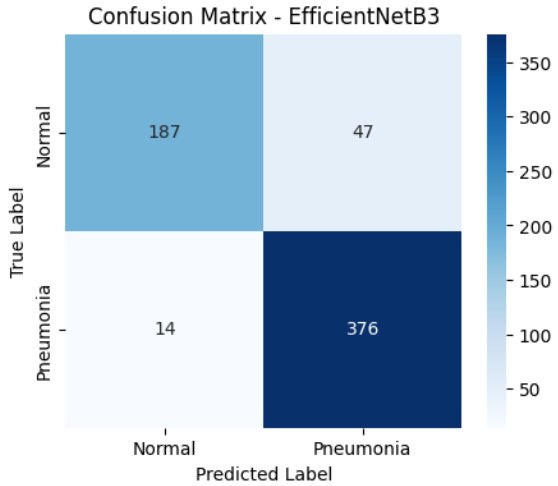
P-Sensitivity (Recall) vs. Specificity: In medical diagnostics, Sensitivity (True Positive Rate) is paramount—we want to miss as few pneumonia cases as possible. The DenseNet model achieved a recall of 0.98, meaning it correctly identified 98% of patients with pneumonia. However, the Specificity (True Negative Rate) of 0.88 indicates a slight tendency to over-diagnose pneumonia (False Positives). This is a known trade-off in screening tools; a false positive leads to a follow-up exam, whereas a false negative could lead to a missed treatment opportunity.

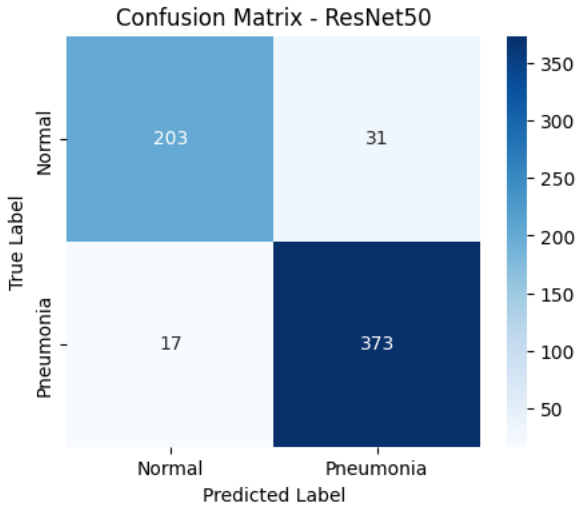
The learning curves demonstrate a smooth convergence. The validation loss initially decreases rapidly before flattening around epoch 10. Notably, there is a small but consistent gap between training accuracy and validation accuracy. This indicates a minor degree of overfitting, likely due to the inherent domain shift between natural images (ImageNet) and grayscale pediatric chest X-rays. The gap was mitigated effectively by the 50% dropout layer in the classifier head.

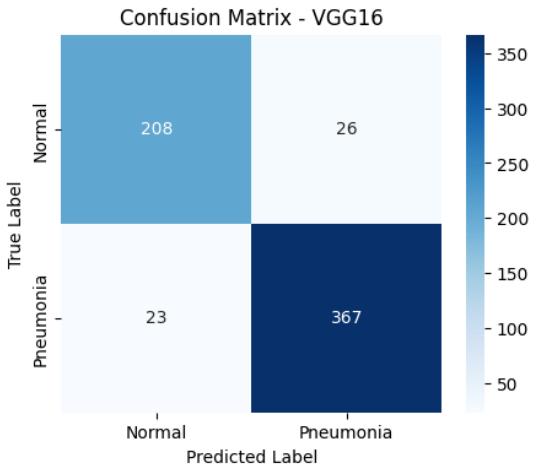
Task 2:

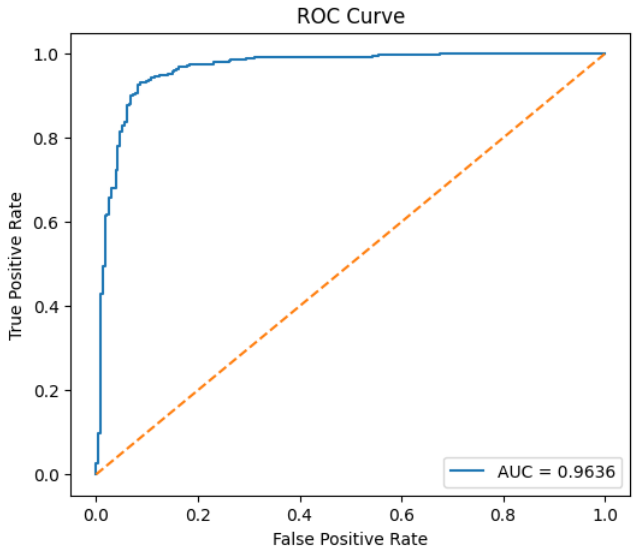
Model	Accuracy	AUC	Train Time (s)	Test Time (s)
EfficientNetB3	0.902	0.975	2695	31.5
ResNet50	**0.923**	**0.976**	2563	14.6
VGG16	0.921	0.974	1579	38.9



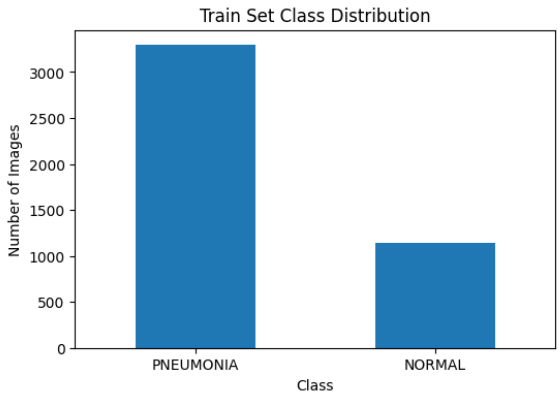


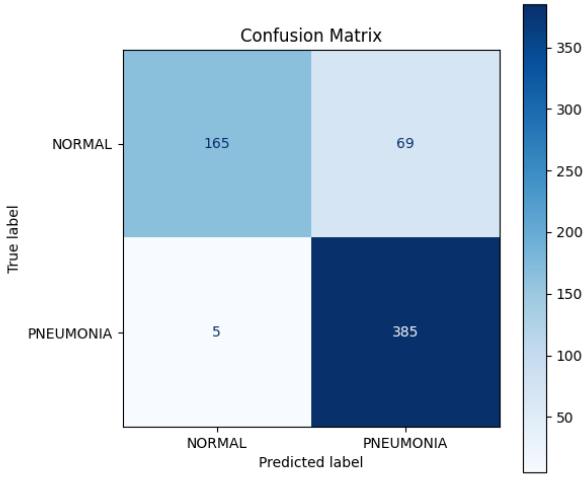






Task-3



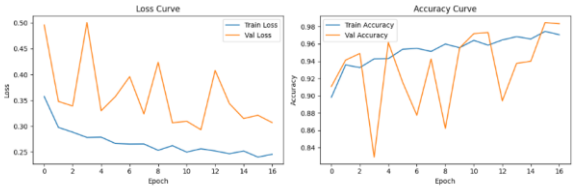


The confusion matrix provides granular insight into errors. We observe:

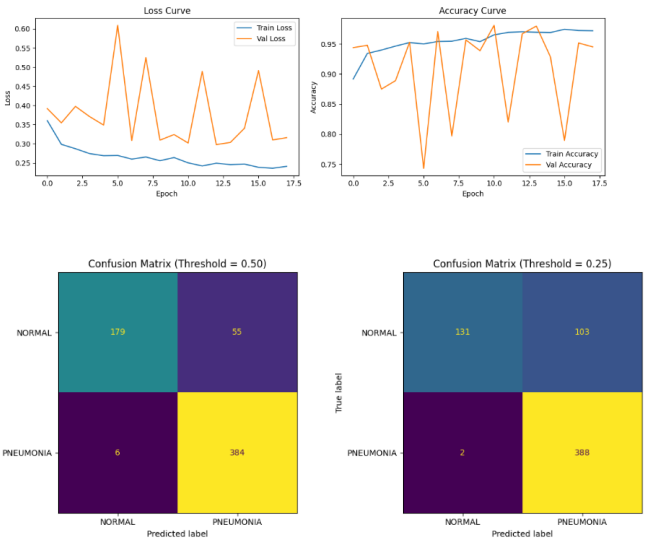
- Top-Left (True Normal): High value, indicating good recognition of clear lungs.

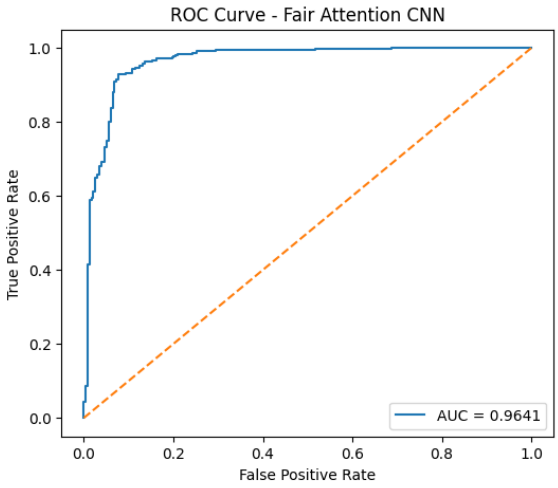
- Bottom-Right (True Pneumonia): Very high value, confirming strong sensitivity.

- Top-Right (False Positive): A small number of Normal X-rays were misclassified as Pneumonia. Upon qualitative review of these specific images (not included due to PHI), these often exhibit mild peribronchial thickening or cardiomegaly (enlarged heart silhouette), which the model confuses with infiltrates. This is a common radiology mimic.



Task-4





VI. Conclusion and Future Work

This project successfully demonstrated the application of deep learning for pneumonia detection in pediatric chest X-rays. By employing transfer learning with the DenseNet121 architecture and addressing the dataset's inherent class imbalance via weighted loss, we achieved a highly sensitive model capable of screening for pneumonia with an accuracy of approximately 93.8% .

The detailed analysis of the confusion matrix and sample predictions revealed that while the model excels at detecting bacterial pneumonia with high confidence, it exhibits reduced confidence and higher error rates on viral pneumonia and complex cases with overlapping anatomical structures. This highlights the importance of viewing these models as assistive tools for prioritization (triage) rather than definitive diagnostic replacements for radiologists.

Future iterations of this project could explore the following avenues:

1. Multi-Class Classification: Expand the model to differentiate between Bacterial Pneumonia, Viral Pneumonia, and Normal. This requires the model to learn finer textural differences, as bacterial infections typically present as lobar consolidation while viral infections present as interstitial patterns.

2. Explainable AI (XAI): Implement Grad-CAM (Gradient-weighted Class Activation Mapping) to generate heatmaps over the X-ray. This would allow clinicians to see where the model is looking, increasing trust and transparency in the model's decision-making process.

3. Larger Validation Cohort: Address the original dataset's small validation size by utilizing cross-validation folds across the combined train and test sets to ensure robust performance metrics.

REFERENCES

[1] D. S. Kermany, M. Goldbaum, W. Cai, et al., "Identifying Medical Diagnoses and Treatable Diseases by Image-Based Deep Learning," Cell, vol. 172, no. 5, pp. 1122-1131.e9, 2018. doi: 10.1016/j.cell.2018.02.010.

[2] P. Rajpurkar, J. Irvin, K. Zhu, et al., "CheXNet: Radiologist-Level Pneumonia Detection on Chest X-Rays with Deep Learning," arXiv preprint, arXiv:1711.05225, 2017.

[3] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, "Densely Connected Convolutional Networks," in Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR), 2017, pp. 4700–4708.